



Hiddenite

△ Emerald-cut gem of the spodumene variety hiddenite

His green variety of the lithium mineral spodumene was discovered by a geologist who had been commissioned by the inventor Thomas Edison to search for any sources of platinum in North Carolina, USA. Mining began in the 1890s at the discovery site, where hiddenite occurred alongside emerald and was, for a time, called “lithia emerald”. Hiddenite crystals are small, seldom exceeding 25mm (1in) in length, and are strongly pleochroic, showing green, bluish-green, and yellowish-green when viewed from different directions.

Specification

Chemical name Lithium aluminosilicate | **Formula** $\text{LiAl}(\text{Si}_2\text{O}_6)$
Colours Green, blue-green, yellow-green | **Structure** Monoclinic
Hardness 6.5–7 | **SG** 3.0–3.2 | **RI** 1.66–1.68 | **Lustre**
 Vitreous | **Streak** Colourless | **Locations** USA, Brazil, China, Madagascar

Elongated crystal | **Rough** | While emerald-green is the preferred colour for hiddenite, lighter green crystals, as here, can also be faceted for jewellery.



Hiddenite in gneiss | **Rough** | Crystals have grown here in a gneiss matrix, although hiddenite is usually found in pegmatites (coarse-grained quartz and feldspar mixes).



Gem-quality crystal | **Rough** | This hiddenite rough is transparent and of a fair green colour. It is likely to be cut into an elongated stone (see right).



Navette cut | **Cut** | Hiddenite crystals tend to be elongated, lending themselves to fancy cuts like this 4.96-carat navette cut with a mix of triangular and rectangular faces.

The tiny settlement of **White Plains, North Carolina**, was renamed **Hiddenite** after the mineral was discovered there



Elongated gem | **Cut** | The clean lines of the facets emphasize the pale blue-green colouring of this rectangular step-cut hiddenite gem.



Faceted rectangle | **Cut** | This 31.60-carat round-cornered rectangular cut stone from Afghanistan displays excellent lustre and clarity, and is unusual for its large size.